

**WYVERN-E: Engineering Design and Experimental Validation of Active Flight Control,
Materials, Aerodynamic Optimization, and Wind Tunnel Evaluation in a
Subscale-Demonstrator Autonomous Guided Rocket**

Allison Hong, Chris Liu, Swaroop K. Sahoo

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Student Information

Student Names: Swaroop Sahoo, Chris Liu, Allison Hong

Project Title: WYVERN-E: Engineering Design and Experimental Validation of Active Flight Control, Materials, Aerodynamic Optimization, and Wind Tunnel Evaluation in a Subscale-Demonstrator Autonomous Guided Rocket

Research Pathway: Engineering/Innovation Design

Intended Majors:

- *Swaroop Sahoo*: Electrical Engineering with Aerospace Engineering Minor
- *Chris Liu*: Aerospace Engineering with Business Minor
- *Allison Hong*: Mechanical Engineering with Aerospace Engineering Minor

Introduction

Active flight control and system guidance in small-scale rocketry requires the simultaneous optimization of aerodynamics, embedded systems, materials science, and control theory. Existing solutions to these problems are confined to proprietary or restricted hardware unavailable for independent research, and no open-source or hobby-scale platform currently integrates dual-mode active control, a distributed avionics stack, and vertical-to-horizontal flight transition capability in a single system. The WYVERN-E is a full-scale prototype autonomous guided rocket built from commercially available components and hobbyist-level manufacturing techniques, designed to fill this gap and produce openly reproducible hardware, datasets, and design methodology for the broader rocketry community. The vehicle employs a hybrid control architecture combining four actively-actuated canard fins (ACFs) and a twin-vane ceramic jetvane thrust-vector control (TVC) system, evaluated across a range of actuator configurations and flight regimes. A distributed three-board avionics stack executes a multi-input PID control loop in C, with attitude estimation running on IMU sensor data. Structural and aerodynamic components are fabricated via FDM using candidate engineering filaments characterized per Dizon et al. (2018), and fin cross-section profiles are selected and evaluated against low-Reynolds-number aerodynamic theory established by Lissaman (1983) and Mueller and DeLaurier (2003). To support ground-based aerodynamic characterization alongside flight testing, a custom open-return low-speed wind tunnel based on the Hofferth (2025) AIAA SciTech modular design is constructed in parallel with the vehicle, following tunnel design criteria from Mehta and Bradshaw (1979) and Pope and Harper (1966). Together, the wind tunnel, computational simulations in OpenRocket and SimFlow, and in-situ powered flight testing form a three-tier experimental framework through which each research question is addressed across multiple validation methods, with all resulting datasets, design files, and firmware released publicly through NAR upon conclusion of the program.

Research Questions

Research Question 1 - TVC vs. ACFs for Efficient Control Authority

Substituting ceramic jetavane thrust-vector control for independently servo-actuated canard fins as the primary attitude actuator, evaluated across a range of airspeeds in both wind tunnel and powered flight conditions, will produce measurable differences in attitude correction gain per degree of actuator input, latency, and achievable pitch and yaw rate, with the objective of determining which actuator type provides greater control authority and whether a crossover point exists between the two systems.

Research Question 2 - Structural Performance in 3D Printable Materials

Holding all FDM print parameters constant across PETG, PETG-CF with 20% carbon fiber reinforcement, PLA Aero, and ASA Aero, this study will characterize and compare each material's flexural stiffness under standardized 3-point bend loading, dimensional deviation from CAD nominal geometry, printed surface roughness, and rate of physical and visual degradation following controlled UV exposure, with the objective of identifying the most capable filament for the airframe and fin fabrication.

Research Question 3 - Aerofoil Profile Efficiency for Subsonic Control

Testing thin symmetric NACA, moderate symmetric NACA, double-wedge, and flat-plate fin cross-sections across incrementally increasing angles of attack in subsonic wind tunnel flow, lift coefficient, drag coefficient, lift-to-drag ratio, and stall onset angle will be compared across profiles to determine which geometry produces the most favorable aerodynamic performance for a fin operating under active deflection.

Research Question 4 - Wind Tunnel Calibration with In-Situ Performance

Incrementally adjusting fan speed and configuration across a range of test section velocity and static pressure settings in the custom open-return low-speed wind tunnel, the Reynolds number similarity ratio, blockage-corrected drag coefficient, and cross-sectional flow uniformity will be evaluated at each condition to identify the combination of flow velocity and static pressure that most closely replicates predicted loading on the vehicle during flight.

Research Question 5 - Accuracy of Flight Control Loop Architectures

Operating proportional-only, proportional-integral, and full proportional integral derivative flight computer control loop configurations with gain values tuned iteratively across a minimum number of repeated flight iterations per configuration, percent deviation from the nominal pre-plotted flight trajectory, lateral displacement from the intended recovery point at parachute deployment, and peak attitude error during the powered flight phase will be quantified and compared to determine which loop architecture produces the greatest trajectory accuracy.

Project Goal

The goal of the WYVERN-E project is to design, fabricate, and flight-test the WYVERN-E system, a recoverable full-scale prototype autonomous guided rocket based on a standard 70mm airframe diameter. The system serves as an integrated test platform for evaluating methods of actively powered flight control, performance of hobbyist and industry-grade 3D printable materials, the quantifiable differences in fin aerofoil geometries, the translation of the constructed test wind tunnel simulations to real world flight data, and the differences between different levels of flight computer control loop architecture. Testing will be conducted utilizing computer-based CFD and FEA simulations, ground-based wind tunnel testing, and in-situ powered flight, with the goal to produce usable datasets and reliable components for use in the the rocketry community.

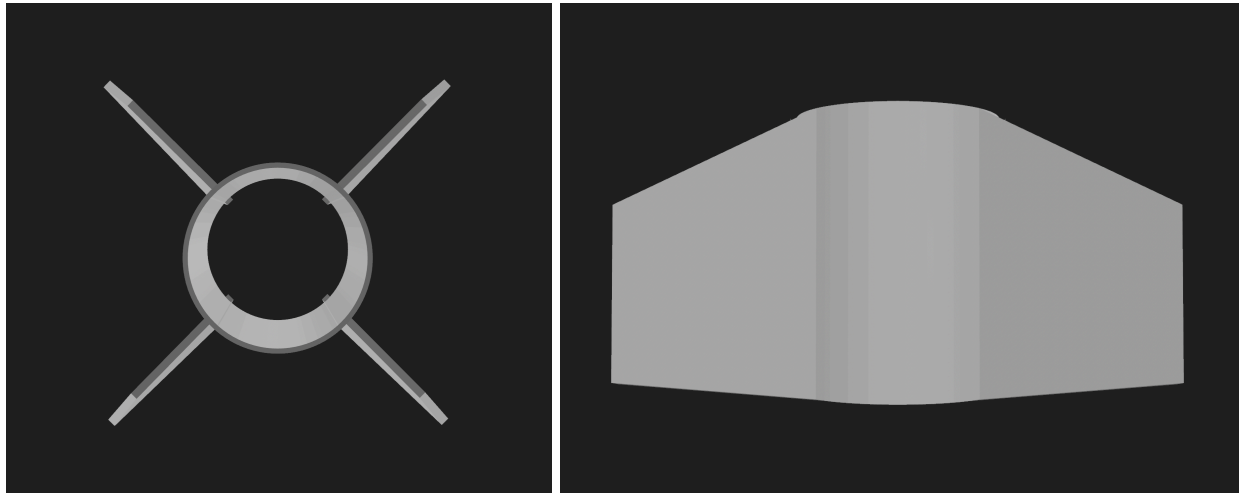
Proposed Methodologies

The WYVERN-E program employs a three-tier experimental framework spanning computational simulation, ground-based wind tunnel testing, and in-situ powered flight testing. Each research question is addressed through at least two of these tiers to enable cross-validation of results. All fabrication utilizes hobbyist-accessible manufacturing methods and commercially available components, consistent with the project's open-source objectives.

Airframe & Materials

All primary structural and aerodynamic components, including the airframe body tube, nose cone, mid-ring fin module, and TVC housing, are fabricated via FDM on a Bambu Lab X1C printer using the factory 0.4 mm brass nozzle. Print parameters including layer height, infill pattern and density, wall count, print temperature, and part cooling are held constant across all test articles to isolate material as the independent variable for Research Question 2, following the methodology established by Popescu et al. (2018) for FDM parameter control. A standardized parameter profile is developed and frozen prior to the first production run, and all airframe prints reference this profile. The four candidate filaments are PETG as a baseline control, PETG-CF with 20% chopped carbon fiber reinforcement, PLA Aero, and ASA Aero. The selection of PETG-CF is motivated by Bhandari et al. (2019), who demonstrated measurable improvements in interlayer tensile strength for short carbon fiber reinforced PETG over unfilled variants. PLA and PETG thermal and mechanical behavior across print conditions is characterized per Hsueh et al. (2021). Dimensional accuracy is verified post-print using digital calipers with deviation from CAD nominal geometry recorded per part, and surface roughness is characterized using profilometry prior to any post-processing. Flexural stiffness of each material is characterized under standardized 3-point bend loading per Dizon et al. (2018), whose mechanical characterization framework for FDM-printed polymers provides the basis for the test protocol used here.

Fins & Aerofoil Profiles



*Fin Profiles shown above as downward, then right view.

Four fin cross-section profiles are fabricated for wind tunnel evaluation: a thin symmetric NACA profile (NACA 0006), a moderate symmetric NACA profile (NACA 0012), a double-wedge, and a flat-plate. Symmetric NACA section aerodynamic data referenced in Abbott and Von Doenhoff (1959) and the NACA 0012 characterization from NACA TN 2502 provide baseline expected performance for the NACA profiles at low angles of attack. All profiles are printed to identical planform dimensions so that only cross-sectional geometry varies between test articles. Low-Reynolds-number aerodynamic behavior of these profile types is framed using Lissaman (1983) and Mueller and DeLaurier (2003), both of which establish that profile geometry effects on lift and drag are particularly pronounced in the subsonic, low-Reynolds-number regime characteristic of the WYVERN-E flight envelope. Canard geometry for the flight vehicle is selected based on wind tunnel results and confirmed through OpenRocket stability simulation using the Barrowman (1967) center-of-pressure method implemented in that tool.

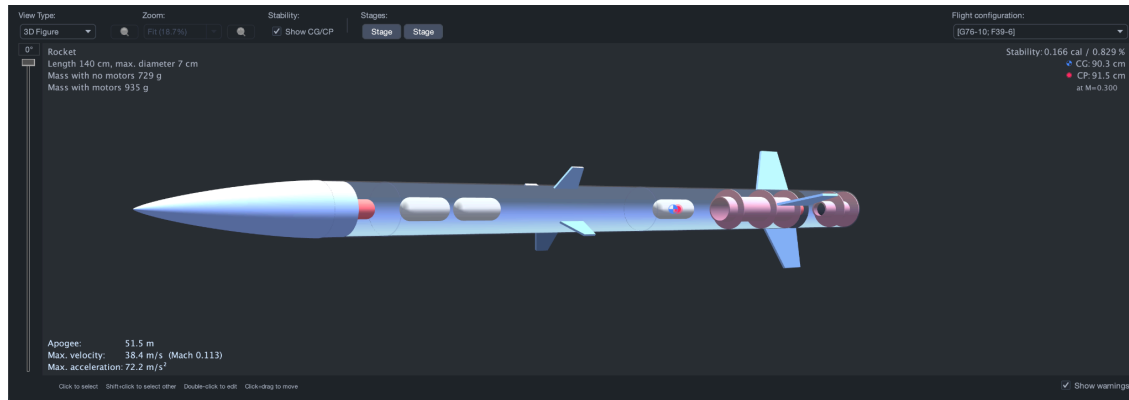
Flight Computer Systems & Avionics

The three-board avionics stack is designed in EasyEDA with trace and via placement handled by the Quilter autorouter, then fabricated boards are manufactured through JLCPCB. The Central Command Module runs on an RP2040 and hosts the primary flight computer firmware. ASAM-1, built around an STM32F411, drives the four KST X08 servos controlling the mid-ring actively controlled fin module, while ASAM-2 uses the same processor to drive three servos controlling the dual ceramic jetvane TVC assembly. The ceramic jetvane material selection follows the thermal and mechanical property specifications outlined in the Precision Ceramics (2021) Macor technical data sheet, which establishes the suitability of machinable glass ceramic for sustained exposure to solid motor exhaust plumes. Jetvane TVC system design and characterization draws on Murty and Chakraborty (2015) for numerical performance

expectations and Liu and Hui (2024) for implementation approaches in micro-scale solid motor TVC systems. Sensor inputs include an ICM-42688-P IMU and BMP388 barometer on the CCM. A Mahony complementary filter (Mahony et al., 2008) runs on IMU data to produce stable attitude quaternion estimates for the control loop input. A LoRa 915 MHz link via the Ebyte E22-900M22S module provides bidirectional telemetry to a laptop ground control station throughout powered flight.

Simulations, Firmware, and Modelling

All flight computer firmware is written in C and developed within the PlatformIO extension for Visual Studio Code, targeting the RP2040 and STM32F411 using their respective PlatformIO frameworks. The firmware implements a multi-input PID control loop that reads IMU attitude quaternion estimates, computes attitude error relative to the pre-loaded nominal trajectory, and outputs servo commands to the ACF and TVC actuators. Three loop configurations are evaluated for Research Question 5: proportional-only, proportional-integral, and full proportional-integral-derivative. Gain values for each configuration are tuned iteratively across repeated flight tests using the Ziegler-Nichols (1942) step-response tuning method. TVC and ACF control architecture design draws on the low-cost launcher guidance framework described by Gordillo et al. (2023) and on NASA TN D-4971 (1968) for fundamental TVC authority and control allocation requirements in solid-propellant systems. OpenRocket is used for pre-flight stability analysis, center-of-pressure and center-of-mass tracking across motor burn, and nominal trajectory prediction per the OpenRocket technical documentation (2023), with simulated outputs compared against flight telemetry as a cross-validation metric.

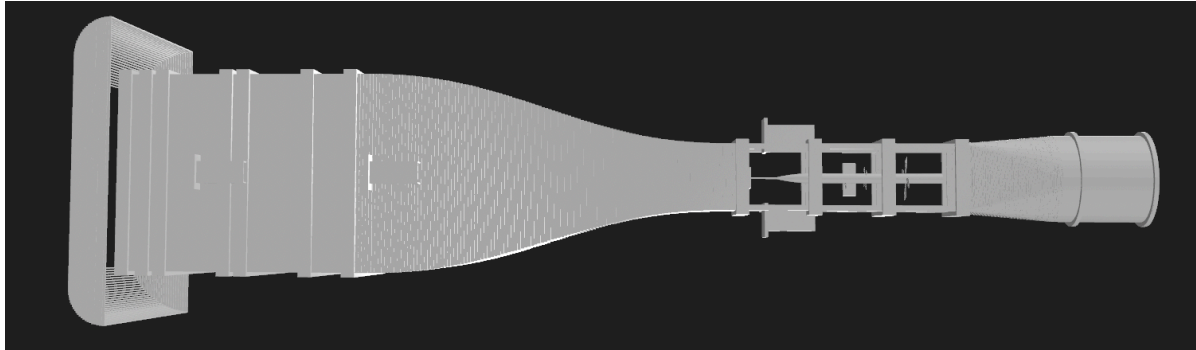


*OpenRocket Model with Stability and Approximate Dimensions

SimFlow, operating on an OpenFOAM finite volume solver, is used for steady-state and transient CFD analysis of individual fin profiles and the full vehicle at representative flight Reynolds numbers consistent with the low-Reynolds-number regime defined by Mueller and DeLaurier (2003) and Lissaman (1983). Simulations are run at incrementally increasing angles of attack corresponding to wind tunnel test conditions, enabling direct comparison of CFD-predicted lift and drag coefficients against measured tunnel values. Mesh refinement

studies are conducted to confirm solution convergence. FEA is conducted on fin and airframe cross-sections for each candidate print material using property inputs derived from the 3-point bend test results, with loading cases corresponding to peak aerodynamic and thrust loads predicted by OpenRocket.

Wind Tunnel



*Initial 3D Model of the low-speed open-return wind tunnel system

The custom open-return wind tunnel is designed per the Hofferth (2025) AIAA SciTech modular configuration and comprises a bell-mouth inlet, a flow-conditioning section with honeycomb and mesh screens, a converging contraction section designed per Bell and Mehta (1988) area ratio and length criteria, an optically accessible test section, and a diffuser leading to a variable-speed fan unit. Tunnel design also draws on the low-speed tunnel design rules of Mehta and Bradshaw (1979) and the reference design methodology of Pope and Harper (1966). Prior to aerodynamic testing, the tunnel is characterized across its full operating range using a Pitot-static probe traversed across the test section cross-section at multiple fan speed settings to map freestream velocity and spatial uniformity, with a uniformity target of less than 2% RMS variation in the core flow region. Blockage corrections are applied to all measured drag coefficients using the Maskell (1963) bluff-body blockage correction method. Reynolds number similarity ratios between tunnel conditions and powered-ascent flight conditions are computed to identify the fan operating point at which tunnel aerodynamic loading on the scaled rocket model most closely replicates predicted loading on the full-scale vehicle during powered ascent.

Each fin profile test article is mounted on a two-axis force balance string within the test section and lift and drag forces are recorded at equal angle-of-attack increments from 0° through post-stall at the calibrated Reynolds-matched condition. Lift coefficient, drag coefficient, lift-to-drag ratio, and stall onset angle are extracted for each profile, with a minimum of three repeated runs per angle per profile conducted to assess measurement repeatability. Mean values from repeated runs are used for all cross-profile comparisons. The complete mid-ring ACF module is tested at representative cruise airspeeds with servo-commanded deflection inputs applied in 2° increments to characterize deflection-normalized lift increment as a function of airspeed, providing the ground-based component of the TVC versus ACF control authority

comparison addressed in Research Question 1. Airfoil section data from Selig (2003) and Selig et al. (1989) are used as reference benchmarks against which tunnel-measured coefficients for the NACA profiles are validated prior to cross-profile comparison.

Flights, Logs, and Data Sharing

All flights are conducted in compliance with NAR (2023) safety code and motor classification standards, with a NAR representative or similarly qualified adult present at each flight session. Comparative flights for Research Question 1 are conducted with the control loop configured to isolate each actuator type: one series uses only the ACF system as the primary attitude actuator and a separate series uses only the TVC ceramic jetvane system. Telemetry logs record actuator command angle, achieved attitude, command-to-response latency, and peak pitch and yaw rates per degree of actuator input across the airspeed range sampled during powered ascent. The crossover airspeed at which one actuator type produces greater control authority gain than the other is identified by comparing these metrics between the two actuator series across the sampled airspeed range. BPS.space TVC design references are used to benchmark expected jetvane authority against prior hobby-scale implementations. For Research Question 5, a minimum number of flight repetitions are completed for each of the three loop configurations, and post-flight telemetry is analyzed for percent deviation from the nominal trajectory, lateral displacement from the intended recovery point at parachute deployment, and peak attitude error during the powered flight phase.

All onboard sensor data is logged at the CCM sample rate and simultaneously downlinked via the LoRa ground station link, with onboard and downlinked logs cross-checked post-flight for completeness and consistency. Upon conclusion of the program, all datasets, CAD files, PCB design files, firmware source code, and OpenRocket and SimFlow simulation files are released publicly through NAR and the project repository in support of the WYVERN-E open-source objectives.

Expected Outcomes



*OpenRocket Render of the WYVERN-E in Flight

We expect to see a few possible outcomes from our research and prototyping journey. We will be able to see the crossover airspeed for which canard fins outperform internal jetvane TVC for control authority and which one provides greater control gain to inform future design decisions, including those for the unexplored transition from vertical to horizontal flight. We will also be able to identify the optimal material for hobbyist manufacturing utilizing 3D printing for rapid prototyping and defining which has the best stiffness, print quality, and overall durability, something not as commonly addressed in rocketry literature which tends to prefer wood or fiberglass construction. In addition we are looking into the aerofoil profiles and will be able to experimentally determine which of them produces the most attitude control in an actively deflected configuration at low Reynolds numbers, with recorded wind tunnel and simulation data being released as an open-source dataset. The development of the WYVERN-E testing wind tunnel will give the rocketry community a cost effective and open-source option for ground-based simulation, powered ascent simulation, and live testing of aerodynamics and control theory. It will also further expand on that control theory by evaluating the complexity of control and the performance regimens that can be achieved with different levels of microprocessor and loop complexity. We hope to be able to release all data as publicly available datasheets through NAR and develop the rocket components into either fledged out products or reproducible systems; still committing to full open-source designs and data sharing.

References

- Abbott, I. H., & Von Doenhoff, A. E. (1959). *Theory of wing sections: Including a summary of airfoil data*. Dover Publications. <https://store.doverpublications.com/0486605868.html>
- Barrowman, J. S. (1967). *The practical calculation of the aerodynamic characteristics of slender finned vehicles* (NASA NTRS accession 20010047838).
<https://ntrs.nasa.gov/citations/20010047838>
- Barrowman, J. S. (1967). *The theoretical prediction of the center of pressure*. Apogee Rockets.
https://www.apogeerockets.com/downloads/barrowman_report.pdf
- Bell, J. H., & Mehta, R. D. (1988). *Contraction design for small low-speed wind tunnels* (NASA CR-182747).
<https://ntrs.nasa.gov/api/citations/19880012661/downloads/19880012661.pdf>
- Bhandari, S., Lopez-Anido, R. A., & Gardner, D. J. (2019). Enhancing the interlayer tensile strength of 3D printed short carbon fiber reinforced PETG. *Composites Part B: Engineering*, 179, 107542. <https://doi.org/10.1016/j.compositesb.2019.107542>
- BPS.space. (n.d.). Thrust vector control. BPS.space. Retrieved April 18, 2026, from
<https://bps.space/products/thrust-vector-control>
- Chacón, J. M., Caminero, M. A., García-Plaza, E., & Núñez, P. J. (2017). Additive manufacturing of PLA structures using fused deposition modelling. *Composite Structures*, 182, 107–116. <https://doi.org/10.1016/j.compstruct.2017.09.004>
- Dizon, J. R. C., Espera, A. H., Chen, Q., & Advincula, R. C. (2018). Mechanical characterization of 3D-printed polymers. *Additive Manufacturing*, 20, 44–67.
<https://doi.org/10.1016/j.addma.2017.12.002>

Hofferth, J. (2025). Modular wind tunnel for STEM education. *AIAA SciTech 2025 Forum*.

<https://doi.org/10.2514/6.2025-2630>

Hsueh, M.-H., Lai, C.-J., Liu, K.-Y., Chung, C.-F., Wang, S.-H., Pan, C.-Y., Huang, W.-C.,

Hsieh, C.-H., & Zeng, Y.-S. (2021). Effects of printing parameters on the thermal and mechanical properties of 3D-printed PLA and PETG. *Polymers*, 13(13), 2092.

<https://doi.org/10.3390/polym13132092>

Lissaman, P. B. S. (1983). Low-Reynolds-number airfoils. *Annual Review of Fluid Mechanics*, 15, 223–239.

<https://bpb-us-w1.wpmucdn.com/sites.usc.edu/dist/4/81/files/2023/05/lissaman-arfm-1983.pdf>

Liu, Y., & Hui, W. (2024). Implementation and verification of a micro-jet-vane system of a solid rocket motor for a micro-nano satellite. *Aerospace*, 11(5), 367.

<https://www.mdpi.com/2226-4310/11/5/367>

Mahony, R., Hamel, T., & Pflimlin, J.-M. (2008). Nonlinear complementary filters on the special orthogonal group. *IEEE Transactions on Automatic Control*, 53(5), 1203–1218.

<https://hal.science/hal-00488376v1/document>

Maskell, E. C. (1963). *A theory of the blockage effects on bluff bodies and stalled wings in a closed wind tunnel* (ARC R&M 3400). Aeronautical Research Council.

<https://reports.aerade.cranfield.ac.uk/handle/1826.2/3452>

Mehta, R. D., & Bradshaw, P. (1979). Design rules for small low speed wind tunnels.

Aeronautical Journal, 83(827), 443–449. <https://doi.org/10.1017/S0001924000031985>

Mueller, T. J., & DeLaurier, J. D. (2003). Aerodynamics of small vehicles. *Annual Review of Fluid Mechanics*, 35, 89–111. <https://doi.org/10.1146/annurev.fluid.35.101101.161102>

- Murty, M. S. R., & Chakraborty, D. (2015). Numerical characterisation of jet-vane based thrust vector control systems. *Defence Science Journal*, 65(4), 263–270.
https://www.researchgate.net/publication/283661759_Numerical_Characterisation_of_Jet-Vane_based_Thrust_Vector_Control_Systems
- National Advisory Committee for Aeronautics. (1951). *Aerodynamic characteristics of NACA 0012 airfoil section at angles of attack from 0° to 180°* (NACA TN 2502).
<https://ntrs.nasa.gov/citations/19930082895>
- National Association of Rocketry. (2023). *NAR safety code and motor classification standards*.
<https://www.nar.org/safety-information/>
- NASA. (1968). *Thrust-vector control requirements for solid-propellant launch vehicles* (NASA TN D-4971).
<https://ntrs.nasa.gov/api/citations/19680019218/downloads/19680019218.pdf>
- OpenRocket Project. (2023). *OpenRocket technical documentation v23.09*.
<https://openrocket.info/documentation.html>
- Pérez Gordillo, A., Simplicio, P., Iannelli, A., & Marcos, A. (2023). Thrust vector control and state estimation architecture for low-cost small-scale launchers. *arXiv*.
<https://arxiv.org/pdf/2303.16983>
- Pillay, S., Vaidya, U. K., & Janowski, G. M. (2009). Effects of moisture and UV exposure on liquid molded carbon fabric reinforced nylon 6 composite laminates. *Composites Science and Technology*, 69(6), 839–846. <https://doi.org/10.1016/j.compscitech.2008.11.012>
- Popescu, D., Zapciu, A., Amza, C., Baci, F., & Marinescu, R. (2018). FDM process parameters influence over the mechanical properties of polymer specimens. *Polymer Testing*, 69, 157–166. <https://doi.org/10.1016/j.polymertesting.2018.05.020>

Pope, A., & Harper, J. J. (1966). *Low-speed wind tunnel testing*. Wiley.

<https://archive.org/details/lowspeedwindtunn0000pope>

Precision Ceramics. (2021). *Macor machinable glass ceramic — technical data sheet*.

https://precision-ceramics.com/wp-content/uploads/2021/06/Macor_Technical_Data_Sheet.pdf

Sahoo, S. (2026, April 11). (Skylight Industries, Ed.). WYVERN PTD Portal; Skylight

Industries. <https://wyvern-e.base44.app/>

Selig, M. S. (2003). *UIUC airfoil data site*. University of Illinois at Urbana-Champaign.

<https://m-selig.ae.illinois.edu/ads.html>

Selig, M. S., Donovan, J. F., & Fraser, D. B. (1989). *Airfoils at low speeds* (Soartech 8).

SoarTech Publications.

https://m-selig.ae.illinois.edu/ads/afplots/airfoils_at_low_speeds.pdf

Ziegler, J. G., & Nichols, N. B. (1942). Optimum settings for automatic controllers. *Transactions of the ASME*, 64, 759–768.

https://davidr.no/iiav3017/papers/Ziegler_Nichols_%201942.pdf

Past Abstracts

Swaroop Sahoo

Evaluating the Efficiency, Practicality, and Effectiveness of Emission/Reception-Capable LiDAR and RaDAR Detection Systems in Autonomous Navigation

Autonomous vehicles are swiftly evolving from experimental prototypes to a social norm. Advanced emitter-based sensor technologies, including LiDAR and RaDAR, are at the heart of these developments. This research has explored, through an analysis of raw performance, which is best for autonomous navigation systems. With an R2 Smart Car, we had extensive testing with the setup to ascertain detection capability across various terrains and conditions. We hypothesized that if an autonomous vehicle employs LiDAR sensors rather than RaDAR sensors, it will realize better detection owing to its higher resolution, precision, and finer performance for short-range distances. During our trials, LiDAR always outperformed RaDAR concerning precision and the recognition of objects within close and medium ranges. However, RaDAR showed better performance in adverse weather conditions, proving that it is more resistant to environmental stressors. Despite LiDAR's superior detection performance, several challenges were identified: higher cost, greater energy demands, and limited field of view. RaDAR, though less effective in performance, claims advantages regarding cost-efficiency, weather robustness, and lower battery requirements. These considerations become critical in balancing efficiency against function while designing a self-driving vehicle. This experiment shows that such a multi-sensor approach is needed to put together the strengths of LiDAR and RaDAR to attain maximum detection accuracy and system reliability. Such integration will be imperative for making robust, cost-effective autonomous systems that are capable of navigating diverse real-world conditions and thereby fostering the feasibility and safety of autonomous vehicles in mainstream society.

Chris Liu

In-Situ resource utilization-derived iron perchlorate redox flow battery for Mars: electrolyte characterization and extreme cold performance validation

Sustained habitation on Mars demands robust energy storage systems capable of reliable operation under extreme cold, especially during night and dust storm periods that render conventional lithium-ion batteries ineffective. This work introduces an in-situ resource utilization (ISRU) strategy for constructing iron perchlorate redox flow batteries, fully leveraging Martian-available materials to achieve extreme cold resilience. Eutectic freezing points and ionic conductivities of three Martian-available electrolytes (iron sulfate, iron chloride, and iron perchlorate) were systematically characterized. Iron perchlorate aqueous solution at 45 wt% displayed a eutectic freezing point of $-78\text{ }^{\circ}\text{C}$, outperforming iron chloride ($-55\text{ }^{\circ}\text{C}$) and iron sulfate ($-10\text{ }^{\circ}\text{C}$). Laboratory-scale single cells were developed via computer-aided design and 3D printing, then tested under simulated Martian low-temperature conditions. The iron perchlorate system maintained 56% of its room-temperature capacity at $-50\text{ }^{\circ}\text{C}$ and remained operational at $-70\text{ }^{\circ}\text{C}$, while iron chloride cells retained only 25% at $-50\text{ }^{\circ}\text{C}$ and lost functionality at lower temperatures. Electrochemical impedance measurements revealed that, although electrolyte resistance increases at lower temperature, charge transfer resistance becomes the dominant limiting factor under extreme cold. The results establish that ISRU-derived iron perchlorate flow batteries offer a feasible, cold-resilient solution for reliable energy storage in future Mars surface operations and settlement, with further performance gains likely through advanced perchlorate brine formulation.

Allison Hong

The Effects of Different Types of Synthetic Retinoids on Bacteria Inhibition: Antibacterial Properties

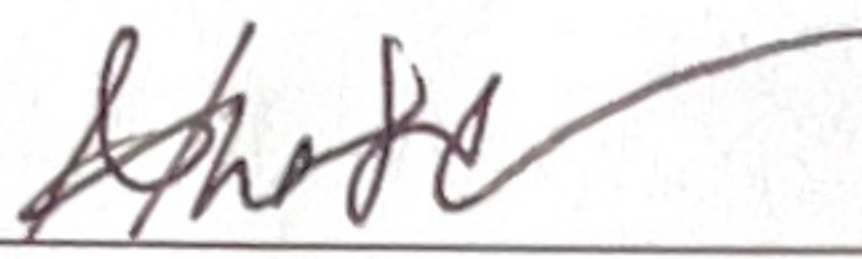
Retinoids are a group of compounds including naturally occurring and synthetic vitamin A metabolites and analogs. The two most common synthetic vitamin derivatives are Tretinoin and Adapalene. This experiment aims to determine the inhibitory properties of Adapalene and Tretinoin on Escherichia Coli. K-12. We hypothesized that the tretinoin group would have a greater inhibitory effect than the adapalene group. We created agar petri dishes and swabbed E. coli. The dishes were incubated for six days in total. Adapalene Gel 0.1% and Tretinoin Gel 0.1% were measured and diluted with Methyl Alcohol. The solutions were added on day two of incubation. The bacterial growth was counted after the incubation period for the number of colonies formed on each dish. The retinoids, especially adapalene, were found to promote the growth of bacteria instead of inhibiting its growth. The data of all three groups represent a polynomial regression, with adapalene and tretinoin exhibiting an exponential pattern and the control group exhibiting a logarithmic pattern. Each result was evaluated with the statistical analysis of a t-test, and all data was significant and not random. Further research is suggested to better understand the retinoids' effects on bacterial growth and inhibition.

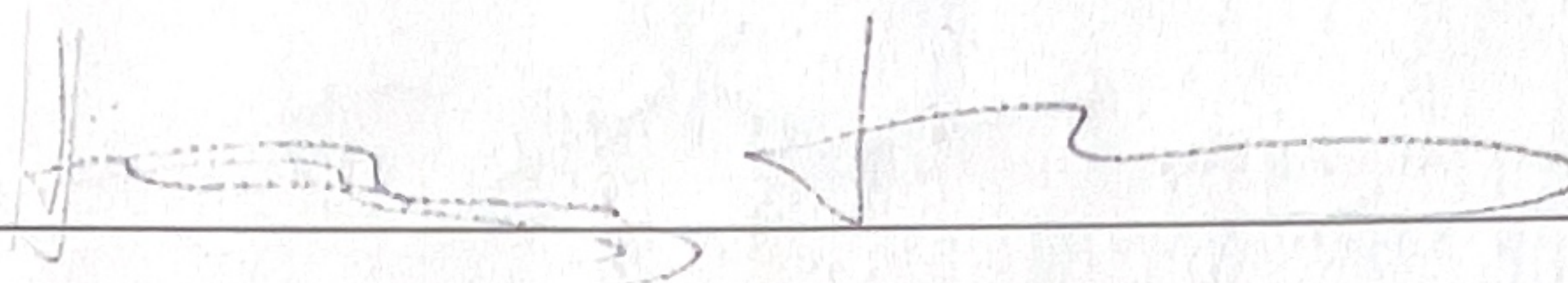
Acknowledgment of Senior Research Form

Acknowledgment Form by **2:10 PM on Monday, April 20, 2026**

We have reviewed the Senior Research requirements and due dates together.

Student's name (please print): Allison Hong

Student's signature: 

Parent's signature: 

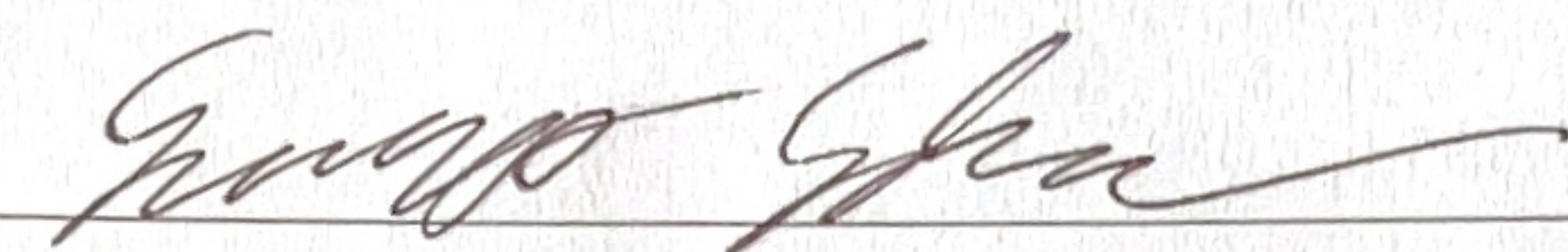
Date: 4/19/2026

Acknowledgment of Senior Research Form

Acknowledgment Form by 2:10 PM on Monday, April 20, 2026

We have reviewed the Senior Research requirements and due dates together.

Student's name (please print): SWAROOP Sahoo

Student's signature: 

Parent's signature: Sujate Sahoo

Date: 04/18/2026

Acknowledgment of Senior Research Form

Acknowledgment Form by 2:10 PM on Monday, April 20, 2026

We have reviewed the Senior Research requirements and due dates together.

Student's name (please print): CHRIS LIU

Student's signature: Chris Liu

Parent's signature: [Signature]

Date: 4/19/26